Mariam Abdullah

Communications Engineer | Data Science & Machine Learning
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PROFESSIONAL SUMMARY

PhD candidate in Electrical Engineering specialising in machine learning, signal processing, and terahertz communication systems, with extensive experience conducting experimental and computational research independently and within multidisciplinary teams. Co-author of 12 peer-reviewed publications (8 journal articles, 4 conference papers) spanning machine learning, wireless communications, and signal processing.

EDUCATION

PhD in Electrical Engineering — Oct. 2022 – Oct. 2026
Adelaide University — *Adelaide, Australia*

BSc in Electrical Engineering — Jun. 2020
American University of Sharjah — *Sharjah, UAE*

RESEARCH EXPERIENCE

PhD Candidate — Oct. 2022 – Present
Terahertz Engineering Laboratory, Adelaide University — *Adelaide, Australia*

- **Research focus:** Deep learning-based signal processing and AI-driven terahertz communication systems.
- **Experimental work:** Built original optical and microwave communication setups in the terahertz channel.
- **Tools and platforms:** MATLAB, Python, PyTorch, and Keras.
- **Publications:** Co-author of 12 publications in machine learning applied to communication and sensing systems (see Publications).

INDUSTRY EXPERIENCE

Data Science Intern — Feb. 2024 – Aug. 2024
Telstra — *Adelaide, Australia*

- Developed deep learning models using **PyTorch** and **Keras** for real-time traffic inference using fibre sensing data.
- Implemented sequential neural network architectures for **object tracking, classification**, and **regression** tasks on **time-series and signal-based datasets**.
- Optimised model performance for real-time inference pipelines in production-like environments.

AI Researcher — Dec. 2021 – Dec. 2022
Datum Consultants — *Dubai, UAE*

- Developed and deployed **deep learning-based computer vision systems** for government applications.
- Built **image classification, detection, and anomaly detection** pipelines using **CNN-based architectures**, implementing end-to-end AI solutions for real-world deployment.

Research and Development Intern — Aug. 2021 – Dec. 2021
Dubai Electricity and Water Authority R&D Center — *Dubai, UAE*

- Developed ML models to predict the properties of chemical compounds using **graph convolutional networks (GCN)**.

JOURNAL PUBLICATIONS

[J1] **M. Abdullah**, M. Larsson, M. Li, E. He, and W. Withayachumnankul, "All-dielectric frequency-diverse terahertz antenna for multi-source direction-of-arrival estimation," *IEEE Trans. Terahertz Sci. Technol.*, pp. 1–13, 2026. doi: 10.1109/TTHZ.2026.3708770

[J2] L. Chen, W. Gao, **M. Abdullah**, J. Hesler, C. Fumeaux, and W. Withayachumnankul, "All-dielectric bandpass filters for IEEE 802.15.3d terahertz communications," *IEEE Trans. Terahertz Sci. Technol.*, vol. 16, no. 5, pp. 564–573, 2026. doi: 10.1109/TTHZ.2025.3647725

[J3] S. Quader, **M. Abdullah**, E. He, C. Fumeaux, and W. Withayachumnankul, "3D-printed frequency-diverse metasurface for camera-based terahertz frequency analyzer," *IEEE Microw. Wireless Technol. Lett.*, vol. 35, no. 12, pp. 2105–2108, 2025. doi: 10.1109/LMWT.2025.3628705

[J4] **M. Abdullah**, J. He, M. Shehata, K. Wang, and W. Withayachumnankul, "Nonlinearity compensation for terahertz communications through neural networks," *Opt. Express*, vol. 33, no. 5, pp. 11178–11191, 2025. doi: 10.1364/OE.541925

[J5] M. S. Li, **M. Abdullah**, J. He, K. Wang, C. Fumeaux, and W. Withayachumnankul, "Frequency-diverse antenna with convolutional neural networks for direction-of-arrival estimation in terahertz communications," *IEEE Trans. Terahertz Sci. Technol.*, vol. 14, no. 3, pp. 354–363, May 2024. doi: 10.1109/TTHZ.2024.3358735

[J6] **M. Abdullah**, K. Chellappan Lethesh, A. A. B. Baloch, and M. O. Bamgbopa, "Comparison of molecular and structural features towards prediction of ionic liquid ionic conductivity for electrochemical applications," *J. Mol. Liq.*, vol. 368, p. 120620, Dec. 2022. doi: 10.1016/j.molliq.2022.120620

[J7] **M. Abdullah**, J. He, and K. Wang, "Weather-aware fiber-wireless traffic prediction using graph convolutional networks," *IEEE Access*, vol. 10, pp. 95908–95918, 2022. doi: 10.1109/ACCESS.2022.3203420

[J8] K. C. Lethesh, A. Bahaa, **M. Abdullah**, M. O. Bamgbopa, and R. A. Susantyoko, "Temperature-dependent electrochemical stability window of bis(trifluoromethanesulfonyl)imide and bis(fluorosulfonyl)imide anion based ionic liquids," *Front. Chem.*, vol. 10, Jun. 2022. doi: 10.3389/fchem.2022.859304

CONFERENCE PUBLICATIONS

[C1] **M. Abdullah**, M. Larsson, M. S. Li, and W. Withayachumnankul, "Multisource direction of arrival estimation in terahertz communications using convolutional neural networks," in *Proc. 50th Int. Conf. Infrared, Millimeter, and Terahertz Waves (IRMMW-THz)*, 2025, pp. 1–2. doi: 10.1109/IRMMW-THz61557.2025.11319767

[C2] **M. Abdullah**, E. He, M. Shehata, K. Wang, and W. Withayachumnankul, "Machine learning for multi-modulation classification in terahertz communications," in *Proc. 50th Int. Conf. Infrared, Millimeter, and Terahertz Waves (IRMMW-THz)*, 2025, pp. 1–2. doi: 10.1109/IRMMW-THz61557.2025.11320121

[C3] **M. Abdullah**, M. S. Li, J. He, K. Wang, C. Fumeaux, and W. Withayachumnankul, "Direction of arrival estimation in terahertz communications using convolutional neural networks," in *Proc. 49th Int. Conf. Infrared, Millimeter, and Terahertz Waves (IRMMW-THz)*, 2024, pp. 1–2. doi: 10.1109/IRMMW-THz60956.2024.10697592

[C4] **M. Abdullah**, J. He, K. Wang, and W. Withayachumnankul, "Improving signal quality in terahertz communications with neural networks," in *Proc. 28th Asia-Pacific Conf. Communications (APCC)*, 2023, pp. 120–125. doi: 10.1109/APCC60132.2023.10460748

HONOURS AND AWARDS

Sept. 2024	Women in STEM Careers and Entrepreneurship Masterclass Travel Grant
Oct. 2023	2023 IEEE Communications Society Student Competition, Honorary Mention
Oct. 2022	University of Adelaide Research Scholarship
Sept. 2016	Al Ghurair STEM Scholarship

TECHNICAL SKILLS

Programming	Python, MATLAB, C++, SQL, HTML
ML frameworks	PyTorch, Keras
Design & test	ADS, PSpice, LabVIEW
Other	Microsoft Office